

30-30
Houn
Lec 67

Lec 67 72

24 A
24 B

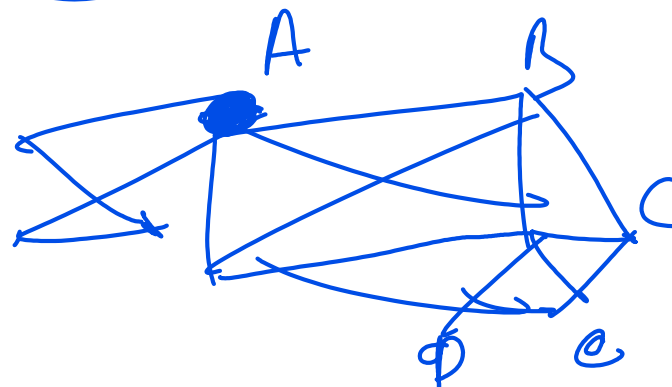
Lec 68

Dijkstra's Algorithm

graph

MST

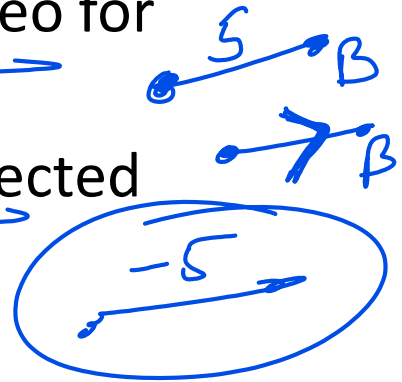
MST - fin $\Rightarrow$ 1-2
100.1



graph Algo 4-5 hours

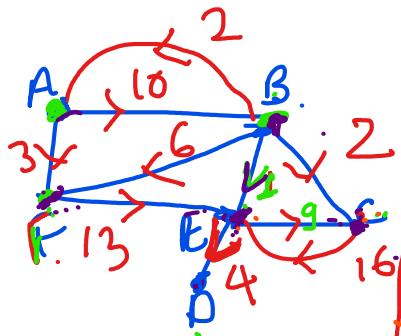
Use: To find Shortest Paths from Source to all Vertices (SSSP)

- Time complexity is same as Prim's Algorithm, Watch that video for more clarifications
- Dijkstra's algorithm can be used on both directed and undirected graphs
- The given graph should not contain any negative edge
- Let's understand the working



Let's

Ex1:



AB = 10
BA = 2
let
Source = A
Delhi = A

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>	<u>E</u>	<u>F</u>
A.	<u>0</u>	∞	∞	∞	∞	∞
F.		10 A	∞	∞	∞	<u>3</u> A
B		<u>10</u> A	∞	∞	13 F	
E		2 B	∞		<u>1</u> B	
C		<u>2</u> B	<u>4</u> E			
D				<u>4</u> E		

